THE LOCALIZATION OF A MORPHISM OF SITES ALONG A MORPHISM OF OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's argument and is explicitly a collection of the identifications available at this point of the development: each of the three assertions is reduced to a named result or to a displayed formal identity, and the author's words describing it as such are preserved verbatim inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The localization of a morphism of sites. Same chapter, Lemma lemma-localize-morphism. Supplies the morphism between the localized sites and its description.
- Relocalizing a site. Same chapter, Lemma lemma-relocalize and Lemma lemma-relocalize-explicit. Supply the composite factorization of the morphism and the explicit description of the pullback.
- The pullback of a representable sheaf. Same chapter, Lemma lemma-pullback-representable-sheaf. Supplies the identifications used in the formal identity of fibre products.

Source reference. Original lemma 03CF: sources/stacks-sites/pinned-original-source.tex, source lines 6601–6601. Navigation pointer invoked at proof line 6601; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EH: sources/stacks-sites/pinned-original-source.tex, source lines 6681–6681. Navigation pointer invoked at proof line 6681; the counted span contains the pointer only and the cited result is not reproduced inside it, so

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Source reference. Original lemma 00Y1: sources/stacks-sites/pinned-original-source.tex, source lines 6685–6685. Navigation pointer invoked at proof line 6685; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 04IK: sources/stacks-sites/pinned-original-source.tex, source lines 6717–6717. Navigation pointer invoked at proof line 6717; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04IL: sources/stacks-sites/pinned-original-source.tex, source lines 6721–6721. Navigation pointer invoked at proof line 6721; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04D3: sources/stacks-sites/pinned-original-source.tex, source lines 6734–6734. Navigation pointer invoked at proof line 6734; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : \mathcal{C} \to \mathcal{D}$ be a morphism of sites corresponding to the continuous functor $u : \mathcal{D} \to \mathcal{C}$. Let $V \in \mathrm{Ob}(\mathcal{D})$, $U \in \mathrm{Ob}(\mathcal{C})$ and $c : U \to u(V)$ a morphism of $\mathcal{C}$. There exists a commutative diagram of topoi*

$$\begin{array}{ccc} Sh(\mathcal{C}/U) & \xrightarrow[j_U]{} & Sh(\mathcal{C}) \\ {\scriptstyle f_c}\downarrow & & \downarrow{\scriptstyle f} \\ Sh(\mathcal{D}/V) & \xrightarrow{j_V} & Sh(\mathcal{D}). \end{array}$$

We have $f_c = f' \circ j_{U/u(V)}$ where $f' : Sh(\mathcal{C}/u(V)) \to Sh(\mathcal{D}/V)$ is as in Lemma 03CF and $j_{U/u(V)} : Sh(\mathcal{C}/U) \to Sh(\mathcal{C}/u(V))$ is as in Lemma 03EH. Using the identifications $Sh(\mathcal{C}/U) = Sh(\mathcal{C})/h_U^\#$ and $Sh(\mathcal{D}/V) = Sh(\mathcal{D})/h_V^\#$ of Lemma 00Y1 the functor $(f_c)^{-1}$ is described by the rule

$$(f_c)^{-1}(\mathcal{H} \xrightarrow{\varphi} h_V^\#) = (f^{-1}\mathcal{H} \times_{f^{-1}\varphi, h^\#_{u(V)}, c} h_U^\# \to h_U^\#).$$

Finally, given any morphisms $b : V' \to V$, $a : U' \to U$ and $c' : U' \to u(V')$ such that

$$\begin{array}{ccc} U' & \xrightarrow[c']{} & u(V') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle u(b)} \\ U & \xrightarrow{c} & u(V) \end{array}$$

commutes, then the diagram

$$\begin{array}{ccc} Sh(\mathcal{C}/U') & \xrightarrow[j_{U'/U}]{} & Sh(\mathcal{C}/U) \\ {\scriptstyle f_{c'}}\downarrow & & \downarrow{\scriptstyle f_c} \\ Sh(\mathcal{D}/V') & \xrightarrow{j_{V'/V}} & Sh(\mathcal{D}/V). \end{array}$$

commutes.

Proof. This lemma proves itself, and is more a collection of things we know at this stage of the development of theory. For example the commutativity of the first square follows from the commutativity of Diagram (04IK) and the commutativity of the diagram in Lemma 03CF. The description of f_c^{-1} follows on combining Lemma 04IL with Lemma 03CF. The commutativity of the last square then follows from the equality

$$f^{-1}\mathcal{H} \times_{h^\#_{u(V)},c} h^\#_U \times_{h^\#_U} h^\#_{U'} = f^{-1}(\mathcal{H} \times_{h^\#_V} h^\#_{V'}) \times_{h^\#_{u(V'),c'}} h^\#_{U'}$$

which is formal using that $f^{-1}h^\#_V = h^\#_{u(V)}$ and $f^{-1}h^\#_{V'} = h^\#_{u(V')}$, see Lemma 04D3. □